

TECHNICAL CASE STUDY & DATA LOG

Dates Company Performance Optimization Project Full-Cycle Data Engineering & Business Intelligence Pipeline

1. Executive Summary & Strategy

Upon initial inspection of the raw enterprise database, severe data integrity violations, anomalies, and architectural errors were discovered across all transaction logs and master sheets. Given that the relational master dimension tables were relatively lightweight (under 1,250 rows), a strategic decision was made to execute a robust Hybrid Data Engineering Pipeline:

- **Master Data Management (MDM):** Extracted, audited, and sanitized dimension/master tables inside MS Excel to establish clean reference files.
- **Relational Database Engine Management (SQL Server):** Loaded the heavy, unstructured transactional logs (sales_dirty) into SQL Server to execute advanced analytical joins, common table expressions (CTEs), and isolate reporting logic via centralized Views.

2. Phase 1: Dimension Tables Sanitization (MS Excel)

2.1 'Customers_Master' Table Tuning

- **Structural Header Alignment:** The initial raw extraction failed to map structural headers properly. Transposed Row 1 to act as the official table column header.
- **Geographic Missing Value Resolution (Country):** Discovered missing Null fields. Initially flagged as 'Unknown' to preserve relational continuity. Through systematic cross-examination of active shipping channels and master index rows, mapped Client C05 strictly to Saudi Arabia based on historic fulfillment corridors.
- **Business-Logic Segmentation Imputation (Segment):** Identified missing Null indicators in customer segments. Conducted a distribution test on historic order sizes and isolated a concrete operational pattern: orders with quantities ≥ 300 KG consistently represented Wholesale behavior, while smaller weights mapped to Retail. Based on this validation rule, the missing value was dynamically imputed to Wholesale.
- **Data De-duplication:** Identified and purged completely redundant duplicate rows for Customer ID C03 to eliminate synthetic database inflation.

2.2 'Inventory_Status' Table Corrections

- **Negative Balance Resolution:** Discovered a critical negative stock balance (Stock_On_Hand_KG = -200) for Ajwa dates. Traced the root cause back to a bulk transaction entry error (-500 KG). Cross-referenced internal group communications which confirmed it was a positive return voucher (+500 KG). Adjusted the ledger balance logically to reflect a positive active shelf availability of 200 KG.

- **Feature Engineering:** Formulated an operational column to reflect true commercial availability using the expression: Available Stock = Stock On Hand - Wastage KG.

2.3 'Products_Catalog' & 'Sales_Reps' Standardization

- **Text Mining & Normalization:** The Product Category attribute suffered from chaotic, mixed-case strings (e.g., cAplItAI / sMaLI). Standardized formatting using proper case logic to ensure uniform naming conventions.
- **Margin Feature Engineering:** Injected a calculated field ($\text{Profit_per_KG} = \text{Price_per_KG} - \text{Cost_per_KG}$) to instantly flag product category base profitability.
- **Context-Aware Quota Imputation:** Identified missing sales target configurations (Null values). Avoided generic flat averages; instead, applied Localized Context Imputation by fetching the specific median target of peer representatives working within the exact same economic zone (The Gulf), yielding an accurate target value of 155,000 EGP.

3. Phase 2: Transactional Data Engineering (SQL Server)

When importing the final dense transactional tables into SQL Server, data-type compiler conflicts emerged. Resolved system boundaries by upgrading the quantity field from SMALLINT to a standard INT descriptor to safely handle system outlier values (such as the 99999 system flag) without crashing. Foreign key dimensions were standardized to NVARCHAR(50).

3.1 Referential Integrity Constraints & Orphan Cleansing

Purged corrupted ghost transactional keys (e.g., Customer C99, Product P99) that did not exist in master lookup tables to secure calculation parameters:

```
DELETE FROM sales_dirty
WHERE Customer_ID = 'C99' OR Product_ID >= 'P99'
      OR Customer_ID >= 'C16' OR Product_ID >= 'P06' OR Rep_ID >= 'R09';
```

3.2 Missing Timestamps & Outliers Removal

Isolated 10 entries entirely missing transaction dates. Because seasonal trends are critical, these were safely eliminated along with severe shipping cost anomalies (-150 EGP) and data entry overflow flags:

```
DELETE FROM sales_dirty WHERE Date IS NULL;
DELETE FROM sales_dirty WHERE Qty_KG = 99999 OR Qty_KG = -500;
DELETE FROM sales_dirty WHERE Shipping_Cost = -150;
```

3.3 Advanced Imputation Using Common Table Expressions (CTEs)

To protect 24 records with missing data weights and 15 records with empty freight configurations from being dropped, engineered grouped calculation loops to fill missing values using isolated group medians:

```
-- Imputing missing quantities via Product Averages
WITH ProductAvgQty AS (
    SELECT Product_ID, AVG(Qty_KG) AS Avg_Qty
    FROM sales_dirty WHERE Qty_KG IS NOT NULL GROUP BY Product_ID
```

```

)
UPDATE s SET s.Qty_KG = a.Avg_Qty
FROM sales_dirty s JOIN ProductAvgQty a ON s.Product_ID = a.Product_ID
WHERE s.Qty_KG IS NULL;

-- Imputing missing shipping costs via Customer Historic Medians
WITH CustomerAvgShipping AS (
    SELECT Customer_ID, AVG(Shipping_Cost) AS Avg_Ship
    FROM sales_dirty WHERE Shipping_Cost IS NOT NULL GROUP BY Customer_ID
)
UPDATE s SET s.Shipping_Cost = c.Avg_Ship
FROM sales_dirty s JOIN CustomerAvgShipping c ON s.Customer_ID = c.Customer_ID
WHERE s.Shipping_Cost IS NULL;

```

4. Analytical Queries & Strategic Insights

4.1 Customer Profitability Mapping

```

SELECT c.Customer_Name,
       SUM((s.Qty_KG * p.Price_Per_KG * (1 - s.Discount_Pct)) - ((s.Qty_KG *
p.Cost_Per_KG) + s.Shipping_Cost)) AS Total_Profit
FROM sales_dirty s JOIN products p ON s.Product_ID = p.Product_ID
JOIN customers c ON s.Customer_ID = c.Customer_ID
GROUP BY c.Customer_Name ORDER BY Total_Profit DESC;

```

💡 BI Key Insight

LuLu Hypermarket emerged as the ultimate financial engine for the enterprise, generating over 4.7M EGP in clean net profits, closely followed by Al Jazeera Markets which locked in the highest standalone profit margin at 42.3%.

4.2 Product Deficit Identification (The Safawi Loss Trap)

```

SELECT p.Product_Name,
       SUM((s.Qty_KG * p.Price_Per_KG * (1 - s.Discount_Pct)) - ((s.Qty_KG *
p.Cost_Per_KG) + s.Shipping_Cost)) AS Sales_Profit
FROM sales_dirty s JOIN products p ON s.Product_ID = p.Product_ID
GROUP BY p.Product_Name ORDER BY Sales_Profit ASC;

```

💡 BI Key Insight

Despite robust transactional numbers, Safawi Dates triggered a severe net loss loop. Cross-referencing the inventory ledger revealed that massive insect infestation cases (costing up to 29,000 EGP per batch in wasted product overhead) completely wiped out its market margins.

4.3 Logistical Freight Cost Outliers (UAE Anomaly)

```

SELECT c.Country, SUM(s.Shipping_Cost) / SUM(s.Qty_KG) AS Shipping_Cost_Per_KG
FROM sales_dirty s JOIN customers c ON s.Customer_ID = c.Customer_ID
GROUP BY c.Country ORDER BY Shipping_Cost_Per_KG DESC;

```

💡 BI Key Insight

Geographical deep-dives uncovered a major freight cost anomaly in the UAE sector. The company absorbs an average of 4.30 EGP per KG shipped to the UAE—almost 4x higher than shipping costs to Saudi Arabia or Oman, leaking vital contract profit margins.

4.4 Markdown Policy Optimization (The 10% Sweet Spot)

💡 BI Key Insight

Data modeling proved that high promotional markdowns (35%) completely cannibalize value. At 35% discount, volume surged, but net invoice profit collapsed to an all-time low of 5,255 EGP. Conversely, a strict 10% discount policy proved to be the optimal 'Sweet Spot', maintaining high demand volumes while maximizing average invoice net profit to a peak of 22,001 EGP.

5. Production Views for Dashboard Deployment

To segregate transactional analytical operations from downstream visualization layer applications (Excel / Power BI), engineered formalized server Views:

```
CREATE VIEW vw_Project_of_Dates AS
SELECT s.Trans_ID, s.Date, c.Customer_Name, c.Country, c.Segment, r.Rep_Name,
r.Region,
    p.Product_Name, p.Category, p.Price_Per_KG, p.Cost_Per_KG, s.Qty_KG,
s.Discount_Pct, s.Shipping_Cost,
    (s.Qty_KG * p.Price_Per_KG * (1 - s.Discount_Pct)) AS Gross_Revenue,
    (s.Qty_KG * p.Cost_Per_KG) AS Product_Cost,
    ((s.Qty_KG * p.Cost_Per_KG) + s.Shipping_Cost) AS Total_Cost,
    ((s.Qty_KG * p.Price_Per_KG * (1 - s.Discount_Pct)) - ((s.Qty_KG *
p.Cost_Per_KG) + s.Shipping_Cost)) AS Net_Profit
FROM sales_dirty s
INNER JOIN products p ON s.Product_ID = p.Product_ID
INNER JOIN customers c ON s.Customer_ID = c.Customer_ID
INNER JOIN sales r ON s.Rep_ID = r.Rep_ID;
```

6. Strategic Action Plan (Executive Brief)

Priority Level	Core Vulnerability	Strategic Executive Recommendation
 CRITICAL	Safawi Inventory Losses & Ajwa Out-of-Stock Risk	Immediately freeze Safawi distributions until warehouse cooling and pest control protocols are audited to stop insect infestation losses.

Urgently restock Ajwa Al-Madinah (current inventory near zero) as it is the company's highest profit generator yielding over 7.7M EGP.

 **HIGH**

Inflatable UAE Freight Costs
& Discount Cannibalization

Renegotiate contract terms with the UAE third-party logistics provider to correct the 4x shipping cost inflation, or pass shipping fees directly to UAE clients. Enforce a strict 10% to 15% backend maximum cap on promotional discounts to protect corporate margins.

 **MEDIUM**

Outdated Sales Quotas &
Severe Ramadan Seasonal
Surges

Re-baseline field sales targets to a minimum of 5M EGP per representative to match true market capacities. Begin aggressive warehouse aggregation and production packing early in December and January to safely absorb the massive volume spikes during the March-May Ramadan peak season.